Jun Zhai

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EDUCATION

National University of Singapore Singapore
PhD Student in Mathematics, NUS Research Scholarship Aug 2024 - Present
Advisor: Prof. Liu Yang, co-advisor: Prof. Xinliang An.

BSc in Mathematics and Physics Aug 2020 - May 2024
Honours with Highest Distinction, Special Programme in Mathematics, Dean's Lists (top 5%).
Honours thesis: Large Cardinals. Advisor: Prof. Chong Chi Tat.

RESEARCH EXPERIENCE

Zhejiang Lab Hangzhou, China
Summer Researcher in AI for Mathematics Summer 2026
Joint NUS and Zhejiang Lab research with Profs. Liu Yang, Lei Zhang and teams from both institutions.

- Core contributor to an AI mathematics agent that solved **at least two open problems in representation theory**, with solutions verified by Prof. Lei Zhang.
- Designed agent workflows and system architecture, integrating the Codex agent harness and MCP tools.
- Developed reusable agent skills and GUI components for reasoning and proof search. Follow-up paper planned.

RESEARCH COMPETITION AND LEADERSHIP

SAIR IGP24 Inverse Galois Problem Challenge
Competition co-organized by Fields Medallist Terence Tao.
Team Lead Summer 2026

- Led the team to **3rd place among 256 participants**, directing mathematical search and compute allocation.
- Collaborated with other teams to **solve the previously open degree-24 existence case of the inverse Galois problem**.
- Built Python multi-agent workflows with prompt design, task scheduling, tool integration and automated evaluation.
- Added parallel computer-algebra verification, CPU-budget controls, SQLite-based experiment reuse and regression tests to support reproducible research.

PAPERS

Zongmin Yu, **Jun Zhai** (co-first author), and Liu Yang. *Agentic Symbolic Search: Characterizing PDEs Beyond Hand-crafted Expressions, Meshes, and Neural Networks.* **Submitted to npj Artificial Intelligence.** arXiv.

ADDITIONAL RESEARCH AND TEACHING

NUS, Teaching Assistant 2023 - Present
Support teaching and assessment in machine learning, calculus, linear algebra and quantitative finance.

Tokyo Institute of Technology, YSEP Research 2023
Pseudospheres and Bäcklund transforms. Supervisor: Prof. Kotaro Yamada.

NUS, Advanced UROPS in Mathematics 2022
Gauss sums and the Weil representation. Supervisor: Prof. Hung Yean Loke.

SKILLS AND HONOURS

Technical: Python, PyTorch, SQL (SQLite), MCP, Codex, agent-system design, reusable agent skills, prompt engineering, tool integration, automated evaluation, regression testing, Lean 4 (Mathlib contributor), Git.
Languages: Chinese (native), English (fluent).
Undergraduate scholarships: Tin Ka Ping Foundation Scholarship and Tham Kai Shui Excelsior Scholarship.